# Sales Forecast

for business planning and supply chain optimization

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# Agenda

1. Problem Statement, Value proposition & Objective
2. Data
3. Prediction model
4. Recommendation & Conclusion

# Problem Statement, Value proposition & Objective

# Problem Statement

Business/Sales planning

Demand forecast

Supply chain optimization

Customer service

**Forecast tool required (Not fortune telling!)**

# Value Proposition

| | | |
|---|---|---|
| Inventory (in days) | 42 days[1)] | 21 days (-50%) |
| Inventory | $55B[2)] | -$13.8B (with 50% probability) |
| Warehousing cost | $182.6B[3)] | -$45.7B (with 50% probability) |

Potential maximum business value up to approximately **$60B/year**
Next 4 weeks sales forecast is required (1 week buffer)

Data as of 2024

1) https://www.gurufocus.com/term/days-inventory/WMT
2) https://www.macrotrends.net/stocks/charts/WMT/walmart/inventory
3) Warehousing cost: Total operation expense ($621.1B[4)]) x Industrial cost average (29.4%[5)]) = $182.6B
4) https://www.macrotrends.net/stocks/charts/WMT/walmart/operating-expenses
5) https://envistacorp.com/blog/toms-takes-walmarts-strategic-investments-grocery-distribution-and-insights-2024-state-of-logistics-report/?utm_source=chatgpt.com

# Objective

- **Objective**
  - Demand forecast ecosystem for next 4 weeks

- **Stakeholders**
  - Business (Sales) planning manager
  - Supply chain manager

# Data

- Data extraction
- Pre-processing
- EDA
- Feature engineering

# Dataset

- Walmart Sales data (US)
  Weekly: 2010-02-05~2012-10-26
  # of lines: 421,570

- Downloaded from Kaggle
  - Sales, Features, Store master
  https://www.kaggle.com/datasets/aslanahmedov/walmart-sales-forecast
  - Holiday: Not covered all holidays
  https://www.kaggle.com/datasets/donnetew/us-holiday-dates-2004-2021

- All data is combined into one dataframe

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 449237 entries, 0 to 449236
Data columns (total 20 columns):
 #   Column        Non-Null Count   Dtype
---  ------        --------------   -----
 0   Date          449237 non-null  datetime64[ns]
 1   Store         449237 non-null  int64
 2   Dept          449237 non-null  int64
 3   Weekly_Sales  449237 non-null  float64
 4   Type          449237 non-null  object
 5   Size          449237 non-null  int64
 6   Temperature   449237 non-null  float64
 7   Fuel_Price    449237 non-null  float64
 8   MarkDown1     158734 non-null  float64
 9   MarkDown2     116712 non-null  float64
 10  MarkDown3     144001 non-null  float64
 11  MarkDown4     140893 non-null  float64
 12  MarkDown5     159573 non-null  float64
 13  CPI           449237 non-null  float64
 14  Unemployment  449237 non-null  float64
 15  year          449237 non-null  int64
 16  quarter       449237 non-null  int64
 17  month         449237 non-null  int64
 18  week          449237 non-null  int64
 19  IsHoliday     107037 non-null  object
```

# Pre-processing

- **Year, Quarter, Month & Week Extraction**

- **Anomaly:**

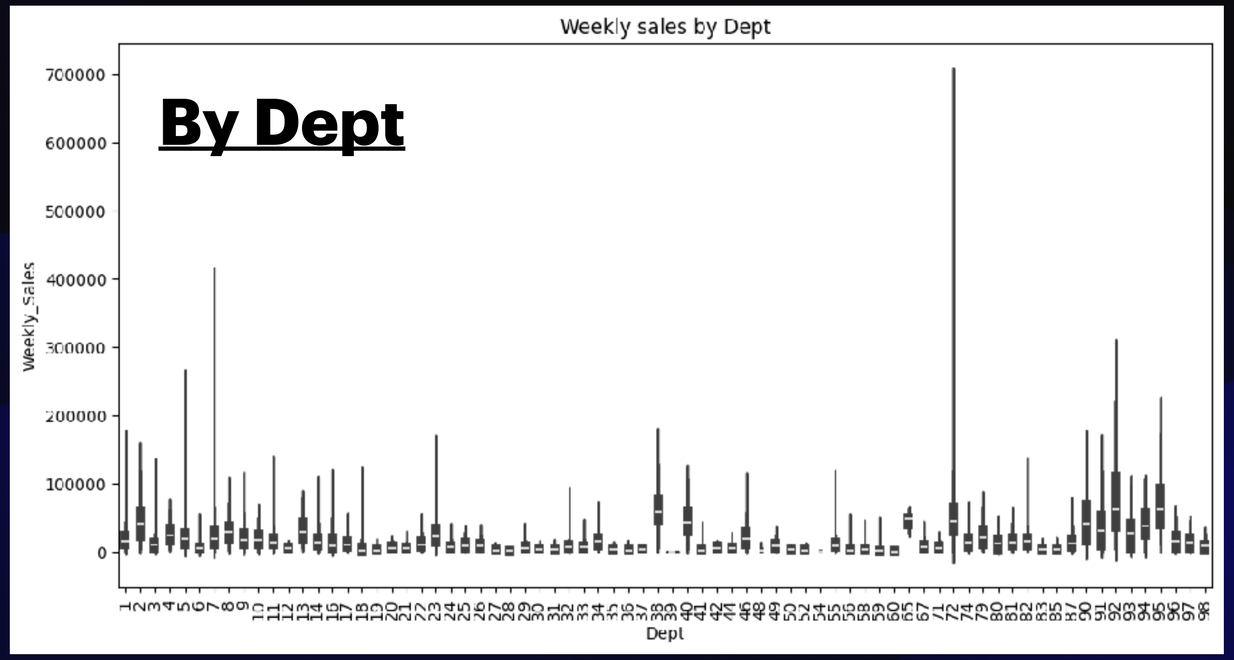
- Irregular sales: No 4-week continuous sales before/ after the forecast creation date → Remove
- Negative value: Markdowns & Weekly sales → Replaced by 0

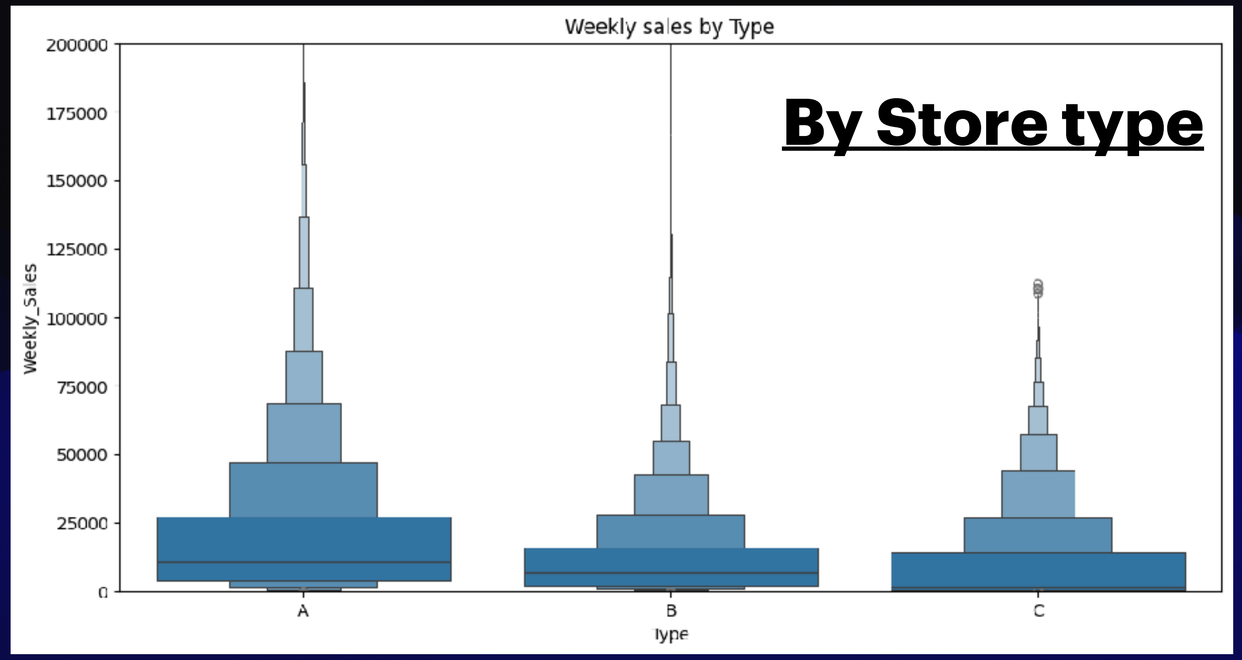
- **Imputation (for lag features):**

- No sales between 2011-04-08 and 2011-05-13 → Imputed 0 between those dates

| Store | Dept | Date | Weekly_Sales |
|---|---|---|---|
| 45 | 47 | 2010-02-26 | -1750.00 |
| 45 | 47 | 2010-05-28 | 100.00 |
| 45 | 47 | 2010-09-24 | 70.00 |
| 45 | 47 | 2010-12-24 | 124.00 |
| 45 | 47 | 2011-02-11 | 0.00 |
| 45 | 47 | 2011-03-25 | 20.00 |
| 45 | 47 | 2011-04-08 | 20.00 |
| 45 | 47 | 2011-05-13 | 20.00 |
| 45 | 47 | 2011-11-18 | -24.00 |
| 45 | 47 | 2011-12-23 | 95.00 |
| 45 | 47 | 2012-08-03 | 224.00 |
| 45 | 47 | 2012-09-07 | 39.96 |

# EDA - Sales by Store & Dept

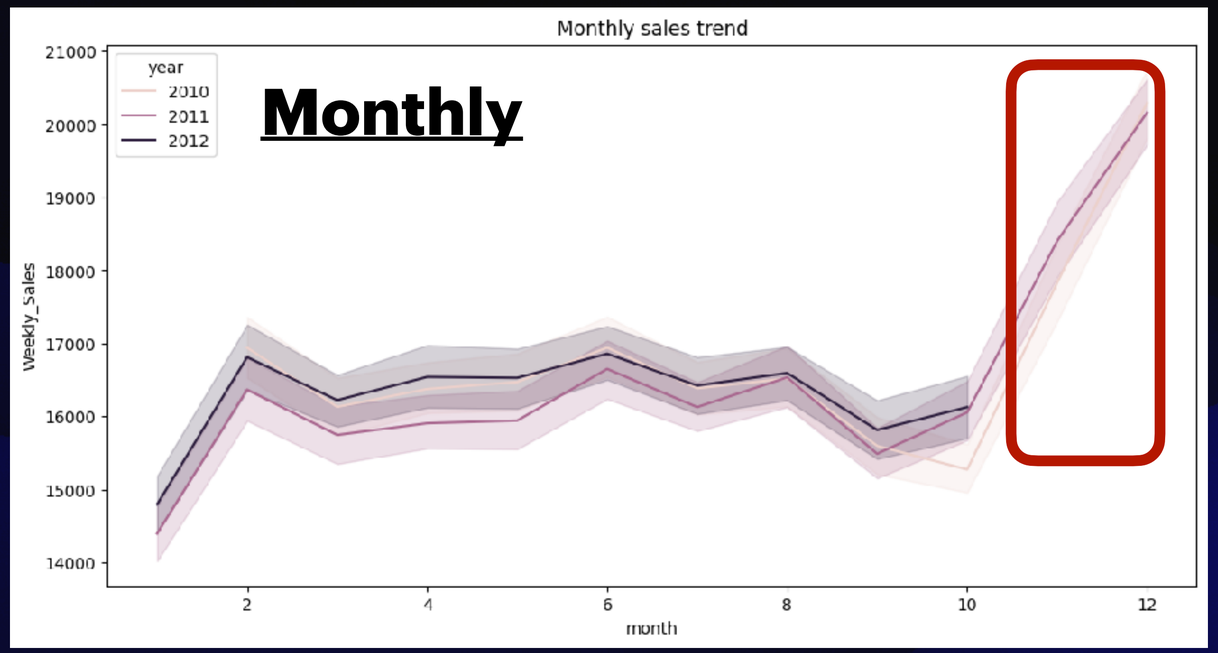


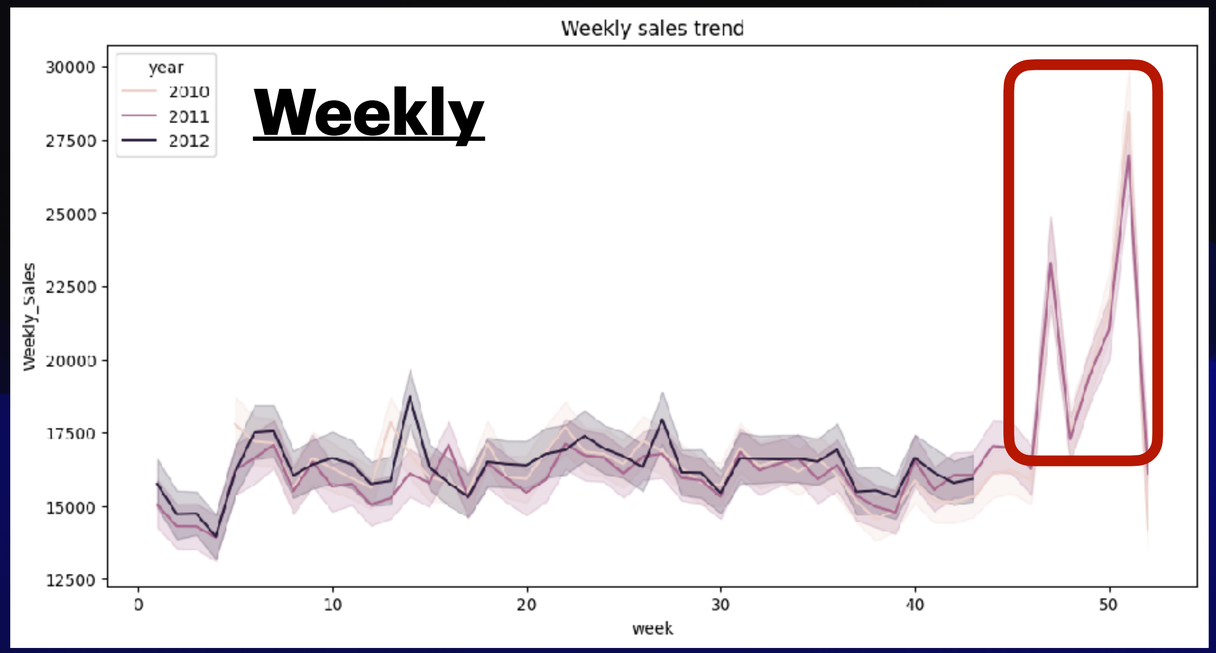




**Store: 45, Type: 3,  Dept: 74**

**Weekly sales have different distribution by store (Type; Size)/Dept**

# EDA - Seasonality

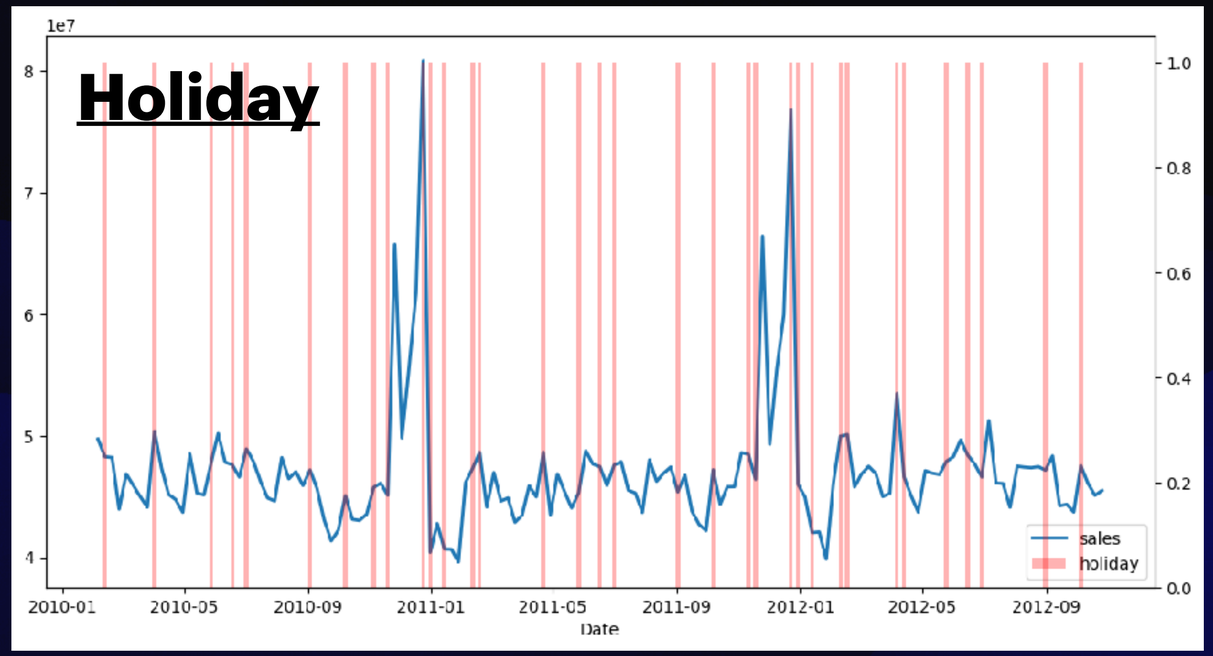


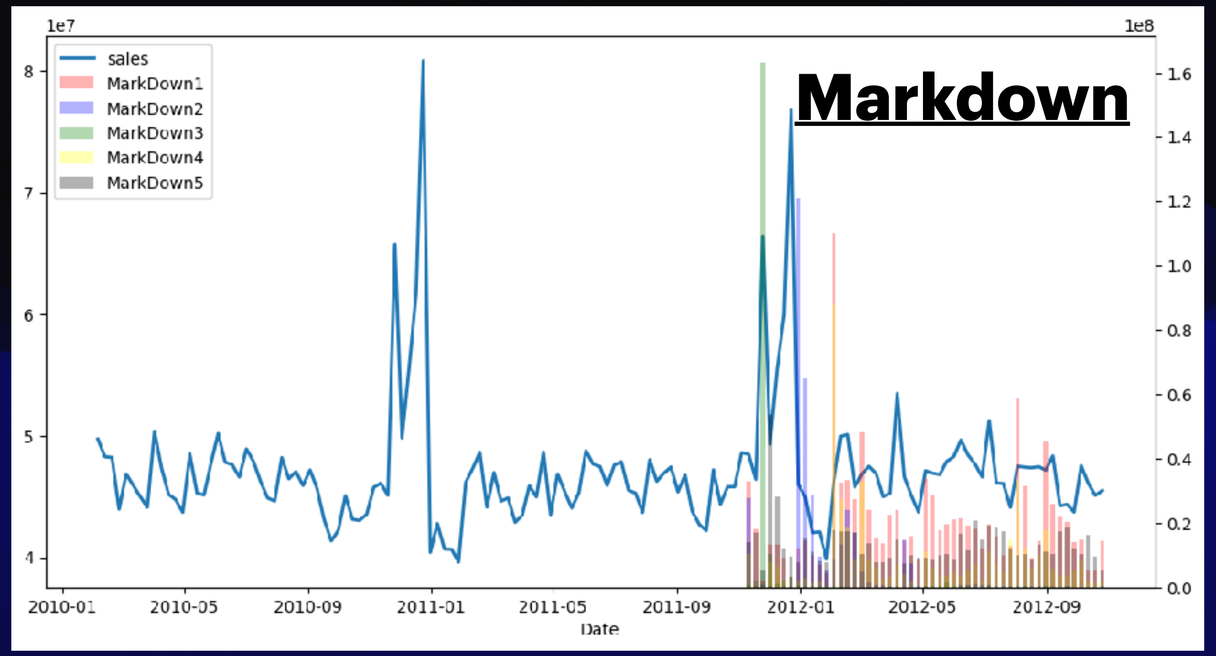




- **Similar trend for every month & Week**

- Higher sales in November & December or on week 48, 50 & 51

# EDA - Effect of Holiday & Markdown




- **Some holidays linked to sales surge; i.e. Christmas**
- **Markdown: only 1 year data → More data required**

# Feature Engineering

- **One-hot encoding: Holidays**
  - Rarely getting new holidays or losing holidays

- **Lag feature: Weekly Sales**
  - Week lag = 1 & 2 introduced
  - ARIMA: Not used & LSTM: Data sequencing

**Lag features**

| Date | Sales | lag1 | lag2 |
|---|---|---|---|
| 1 | 100 | 0 | 0 |
| 2 | 200 | 100 | 0 |
| 3 | 150 | 200 | 100 |
| 4 | 80 | 150 | 200 |

# Prediction model

- **Time series regression models: Multistep time series forecast**

ARIMA, ElasticNet, Random Forest, Light GBM, RNN: LSTM

- **Train-test data split: Time series hold-out**
- **Hyper parameter tuning: GridSearchCV**
- **Model selection**

# Multistep forecast *Except ARIMA

**First step**

| Week | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Input | 100 | 200 | | | |
| Output | | | 180 | | |

**Second step**

| Week | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Input | 100 | 200 | 180 | | |
| Output | | | | 160 | |

**Third step**

| Week | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Input | 100 | 200 | 180 | 160 | |
| Output | | | | | 170 |

## Forecast horizon: 3,4,5

| Week | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Output | | | 180 | 160 | 170 |

# MAPE

Mean Absolute Percentage Error

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right|$$

- Common metric across business, economics, and data science
- Error as a percentage
  → Easier to interpret and communicate the model accuracy
- MAPE is calculated at the lowest granularity: by Store & Dept

# Model evaluation

| Model | Train (MAPE) | Test (MAPE) | Execution time (Train, sec) | Execution time (Test, sec) |
|---|---|---|---|---|
| ARIMA | **12.8%** | **12.3%** | **42** | **9** |
| ElasticNet | **13.5%** | **14.4%** | **132** | **0.3** |
| Random Forest | **3.9%** | **10.4%** | **2423** | **2** |
| LightGBM | **10.0%** | **10.1%** | **5** | **0.1** |
| LSTM | **12.2%** | **16.6%** | **770** | **3** |

**Recommended model: LightGBM**

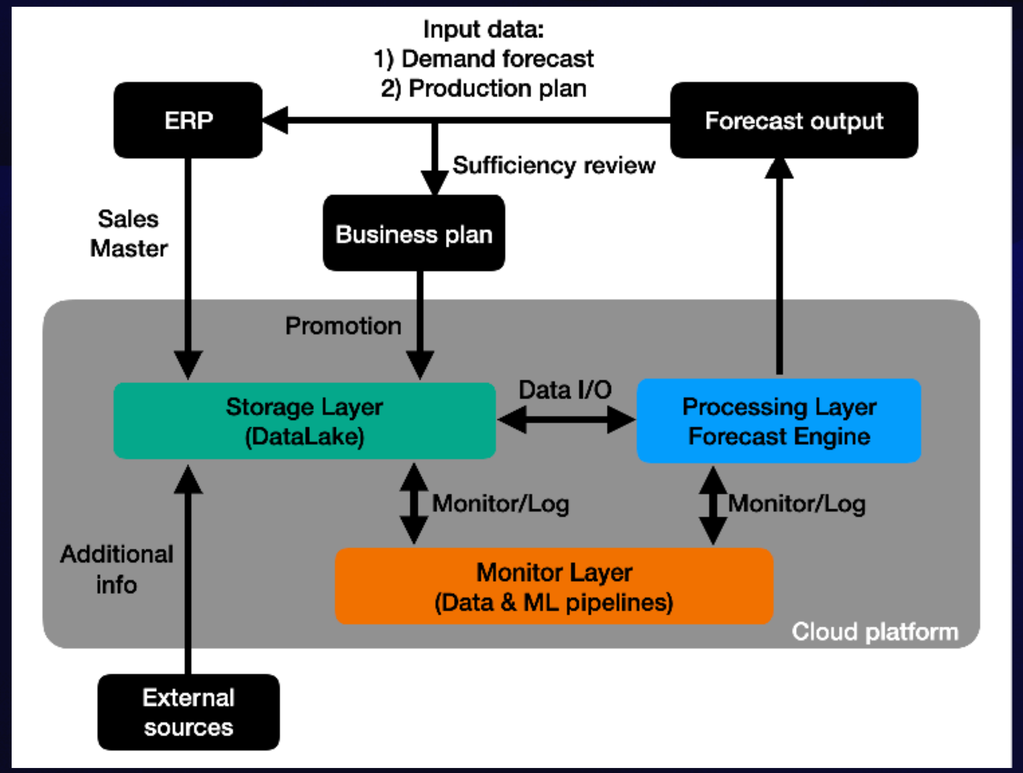
Reliable & scalable MAPE and fast (less computing cost)

# Recommendation

# Recommendation

**Develop a demand forecast ecosystem with Light GBM for upcoming 4-week business & supply chain planning**

Solution diagram




Cost implementation* (Rough estimation)

| Items | Annual cost |
|---|---|
| Cloud service | **$150M** |
| Development resources<br>DS, DE & PM 1FTE | **$600M** |
| Ongoing resources<br>DS+DE 0.4FTE & mgr 0.1FTE | **$100M** |
| Miscellaneous | **$50M** |
| Annual | **$900M** |

**Potentail value = $60B**

*Azure Cost Calculator

# Limitation & Improvement

- **Limitations**

- Computational cost of training is high: Hyper parameters not fully tuned
- Markdown: We only have 1 year of data - If promotion driven store, need more data
- Later horizon (week 3 or 4): MAPE gets worse - Lag features from forecast itself
- No prediction for irregular sales
- Code needs to be refactored

- **Improvement**

- Model tuning: hyper-parameters
- Other features: lag, rolling, tsfresh, Fourier decomposition etc.
- Continuous training is required to capture the latest trends
- Expert’s help: Markdowns or new data & irregular cases

# Conclusion

- More historical Markdown data
- Demand forecast ecosystem with Light GBM for upcoming 4-week business & supply chain planning

# Q&A